

Getting Started with AWS CLI and boto3 for S3

Description

This guide will help you set up your AWS account, configure the AWS CLI, and use `boto3` (the AWS SDK for Python) to interact with Amazon S3. By the end, you'll be able to upload and download files from S3 buckets directly through Python code.

Installation and Setup

1. Create an AWS Account

- Go to [AWS Console](#).
- Sign up for a free account (you'll need a credit card, but you can use the free tier).
- Once signed in, create an **IAM User** (not the root account) for security.
 - Navigate to **IAM > Users > Add Users**.
 - Enable **Programmatic access**.
 - Assign permissions (for simplicity, you can start with `AmazonS3FullAccess`).

2. Install AWS CLI

Install the AWS Command Line Interface (CLI) on your system:

```
# macOS / Linux  
brew install awscli
```

```
# Windows (via Chocolatey)  
choco install awscli
```

3. Verify Installation

```
aws --version
```

4. Configure your AWS Account

```
aws configure
```

1. Make sure to get your access keys from the AWS console.

- You will be prompted for:
 - AWS_ACCESS_KEY_ID
 - AWS_SECRET_ACCESS_KEY
 - REGION
 - OUTPUT_FORMAT

5. Install Python Modules

```
pip install boto3 pandas
```

Basic Usage with S3 Client

1. Create a S3 Client

```
import boto3

# Create an S3 client
s3 = boto3.client("s3")

# List buckets
response = s3.list_buckets()
print("Your Buckets:")
for bucket in response["Buckets"]:
    print(f" {bucket['Name']}")
```

2. Upload a File to S3

```
s3.upload_file("local_file.csv", "my-bucket", "uploads/local_file.csv")  
print("Upload complete!")
```

3. Download a File from S3

```
s3.download_file("my-bucket", "uploads/local_file.csv", "downloaded_file.csv")  
print("Download complete!")
```